Masked Self-Distillation Domain Adaptation for Hyperspectral Image Classification

Zhuoqun Fang, Wenqiang He, Zhaokui Li, *Member, IEEE*, Qian Du, *Fellow, IEEE*, and Qiusheng Chen

***Abstract*— Deep learning-based unsupervised domain adaptation (UDA) has shown potential in cross-scene hyperspectral image (HSI) classification. However, existing methods often experience reduced feature discriminability during domain alignment due to the difficulty of extracting semantic information from unlabeled target domain data. This challenge is exacerbated by ambiguous categories with similar material compositions and the underutilization of target domain samples. To address these issues, we propose a novel masked self-distillation domain adaptation (MSDA) framework, which enhances feature discriminability by integrating masked self-distillation (MSD) into domain adaptation. A class-separable adversarial training (CSAT) module is introduced to prevent misclassification between ambiguous categories by decreasing class correlation. Simultaneously, CSAT reduces the discrepancy between source and target domains through biclassifier adversarial training. Furthermore, the MSD module performs a pretext task on target domain samples to extract class-relevant knowledge. Specifically, MSD enforces consistency between outputs generated from masked target images, where spatial–spectral portions of an HSI patch are randomly obscured, and predictions are produced based on the complete patches by an exponential moving average (EMA) teacher. By minimizing consistency loss, the network learns to associate categorical semantics with unmasked regions. Notably, MSD is tailored for HSI data by preserving the samples' central pixel and the object to be classified, thus maintaining class information. Consequently, MSDA extracts highly discriminative features by improving class separability and learning class-relevant knowledge, ultimately enhancing UDA performance. Experimental results on four datasets demonstrate that MSDA surpasses the existing state-of-the-art UDA methods for HSI classification. The code is available at https://github.com/Li-ZK/MSDA-2024.**

***Index Terms*— Classification, hyperspectral image (HSI), knowledge distillation, masked image modeling (MIM), unsupervised domain adaptation (UDA).**

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I. Introduction

HYPERSPECTRAL images (HSIs), containing dozens or hundreds of narrow spectral bands, are widely utilized in remote sensing applications, such as precision agriculture, environmental monitoring, and mineral exploration. HSI classification [1], [2], [3], [4] is a useful technique in practical applications as it leverages both rich spectral and spatial information for fine land cover classification. Since labeling remote sensing data is difficult and expensive, it is desirable to reuse labeled data in relevant scenes. However, there is often a spectral shift between two HSIs due to variations in acquisition conditions, including different sensors, times, or regions. Consequently, directly applying a model trained from one scene (source domain) to another (target domain) often results in poor performance. To address this issue, various approaches have been investigated, including few-shot learning [5], [6], [7], [8] and transfer learning [9], [10], [11], among which unsupervised domain adaptation (UDA) is the most popular one. The UDA method [12], [13] alleviates spectral shift and classifies target samples using both labeled source data and unlabeled target data.

Conventional UDA methods typically use shallow hand-crafted features for prediction. They can be broadly categorized into three types: instance-based, classifier-based, and feature-based approaches. Instance-based approaches [14], [15], [16] reduce domain shift by reweighting or extracting important instances. Classifier-based approaches [17], [18], [19] focus on training a robust classifier that can generalize well to the target domain. Feature-based approaches, e.g., [20], [21], [22], which are the most commonly used, aim to extract domain-invariant features to minimize interdomain distribution discrepancies in the feature space. Nonetheless, these methods are limited in their ability to adaptively learn features from different data.

Recently, deep learning-based UDA methods have emerged, enabling automatic feature extraction. Harnessing this capability, deep UDA methods commonly employed feature-based approaches for domain alignment and achieved promising performance. One type of this method is distance-based UDA approaches, which minimize the distance metric of the features from different domains to extract domain-invariant features. For example, Long et al. [23] proposed a deep adaptation network (DAN) that extends deep convolutional neural networks to UDA and reduces domain shift by minimizing the maximum mean discrepancy (MMD) distance. In the

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work by Sun and Saenko [24], a deep domain adaptation with correlation alignment (D-CORAL) was proposed, which aligns the covariance matrices of deep features in the two domains, making the learned features more domain-invariant. Li et al. [25] proposed a supervised contrastive learning-based UDA (SCLUDA) for HSI classification, which introduces domain similarity loss and contrastive learning to achieve compact interdomain distribution and enhances data separability within one domain. Another type of deep UDA method is adversarial-based approaches, which introduce adversarial training instead of distance metrics to reduce domain discrepancy. Ganin et al. [26] proposed domain-adversarial neural networks (DANNs) to promote the emergence of features using a domain classifier in an adversarial way. In the work by Deng et al. [27], an Euclidean distance-based deep metric model for UDA (ED-DMM-UDA) was trained to generate a metric value between features and added it to an adversarial framework to adapt feature distributions. Liu et al. [28] presented class-wise distribution adaptation (CDA), which generates domain-invariant features by adopting class-wise MMD and class-wise adversarial training, while also using pseudo-labels of target data in the model optimization process. Fang et al. [29] proposed a confident learning-based domain adaptation (CLDA) for cross-scene HSI classification, which utilizes adversarial training to align distributions and uses confidence learning to select high confident target pseudo-labels to improve the discriminative ability of neural networks. Huang et al. [30] effectively extracted spectral–spatial joint information using a two-branch attention adversarial domain adaptation (TAADA) network. In general, both distance-based and adversarial-based approaches reduce interdomain discrepancy by domain alignment, thus extracting domain-invariant features to improve the target domain classification.

Although domain alignment is a prevalent technique in UDA, it faces limitations. That is because domain alignment may bring about deterioration of feature discriminability when feature transferability is strengthened [31]. This phenomenon is exacerbated by ambiguous categories and underutilization of the target domain. First, some ambiguous categories with similar material compositions are prone to misclassification. For example, in the Houston dataset, "nonresidential buildings" are easily misclassified as "residential buildings" or vice versa. That is because these buildings consist of similar materials with similar spectrums, and there are no labels as supervision during the process of domain alignment. This class confusion problem is seldom considered by current UDA methods in HSI classification. Second, the target domain data is underutilized. The knowledge within the target samples, particularly that relevant to the land cover categories, can help the model extract discriminative features [32]. Nevertheless, extracting such knowledge from unlabeled data remains challenging. On the one hand, due to the spectral shift, the approach of extracting class-relevant knowledge from the source samples is inapplicable to the target samples. On the other hand, domain alignment alone cannot effectively enhance the model's ability to extract knowledge inherent in the unlabeled target samples. Therefore, it is crucial to develop a novel strategy for facilitating the extraction of class-relevant knowledge from the target domain data.

To enhance feature discriminability, several methods have been explored to utilize target domain data. Ning et al. [33] proposed contrastive learning based on category matching (CLCM) for UDA. It introduced contrastive learning to attract the same class samples across domains, while strategically selecting high-confidence target samples with pseudo-labels to boost the learning process. CDA [28] employed class-wise domain alignment guided by pseudo-labeled target samples. Other UDA methods such as CLDA [29] and SCLUDA [25] utilized high-confidence pseudo-labels as supervision signal to directly train the classifier on target samples, thereby improving feature discrimination. However, these methods utilized target domain data at either the class or instance level, heavily relying on the accuracy of pseudo-labels. This reliance hinders their ability to extract discriminative information contained within a target sample and makes them susceptible to label errors.

To overcome the aforementioned limitations, we propose a masked self-distillation domain adaptation (MSDA) method for HSI classification. Unlike existing methods that rely on high-quality pseudo-labels, MSDA leverages class-separable adversarial training (CSAT) and masked self-distillation (MSD) to extract the most discriminative features for accurate target domain prediction. CSAT reduces the misclassification of ambiguous classes in the target domain while promoting domain alignment. MSD fully exploits target domain data to learn class-relevant knowledge. Both components synergistically enhance feature discriminability, thereby improving classification performance.

To address the class confusion problem, CSAT incorporates a minimize class confusion (MCC) loss [34] into a biclassifier adversarial training [35], [36], which leverages two distinct classifiers in an adversarial manner to align marginal distributions across domains. During domain alignment, ambiguous classes may exhibit high correlation, making it challenging for the model to distinguish between them. Consequently, reducing class correlation is crucial to mitigate this issue. To achieve this, the MCC loss estimates the pairwise class correlation based on the model outputs on the target domain. By minimizing this loss, the model can differentiate between ambiguous classes, thereby facilitating the extraction of class-separable features during domain alignment.

Theoretical studies [37], [38] have demonstrated that incorporating a reasonable masking strategy can enhance feature discriminability. Therefore, to fully utilize target domain data, MSD creates a pretext task on target samples by combining self-distillation with masking strategies. In the self-distillation paradigm, an additional teacher model is introduced to improve the performance of the student model. The teacher model is equivalent to an ensemble learning model [39], generating better feature representations than the student model throughout the training process [32], thus guiding the student model's training and gradually improving its feature representations. Furthermore, a novel off-centric spatial–spectral masking strategy is introduced for the target samples, i.e., HSI

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patches. To preserve category information intact, the strategy retains the center pixel, the object to be classified, while masking out a random spatial–spectral selection of the target sample. All masked samples are passed through the student model, while only complete samples are passed through the teacher model. Consequently, the student model must fully utilize and mine the limited information contained in the remaining parts of the masked sample to be consistent with the teacher. Since the masks are random, each local position in the target sample will be fully explored, and those parts that contain class-relevant knowledge will be more attended to by the student model, which is beneficial for extracting class-relevant knowledge.

Compared with recent work in this field, our method has the following three contributions.

1) A new framework combining MSD with CSAT is proposed, which enables the extraction of discriminative domain-invariant features. To the best of our knowledge, this is the first study to integrate MSD into domain adaptation for cross-scene HSI classification.
2) A novel CSAT scheme is introduced that leverages both MCC and biclassifier adversarial training to enhance the feature separability between ambiguous classes with similar material components during domain alignment.
3) An MSD module is specifically designed for unlabeled HSI data, which employs an off-centric spatial–spectral masking strategy to thoroughly exploit the target domain data, and effectively extract class-relevant knowledge from them.

The experimental results demonstrate that our method can outperform the state-of-the-arts in terms of overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) evaluated on five different cross-domain tasks.

The remainder of this article is organized as follows. The related work is introduced in Section II, including self-distillation and masked image modeling (MIM). In Section III, the proposed MSDA method and the training strategy are presented in detail. The experimental results are discussed in Section IV, and the conclusion is drawn in Section V.

II. Related Work

A. Self-Distillation

Knowledge distillation [40], [41], [42] is a learning paradigm where a student model is trained to match the output of a given teacher model so that the student model can be improved by the teacher. Instead of bringing in a pretrained external teacher, self-distillation [39], [43], [44] aims to distill the knowledge in a model itself and use it for training the model. It means that a student model becomes a teacher model itself, which gradually utilizes its own knowledge to soften the hard targets to be more informative during training. In terms of the model update strategy, self-distillation methods such as [39] use a mean teacher model that is derived from the student itself. Specifically, the teacher shares the same structure with the student, while the parameters of the teacher are exponential moving averages (EMAs) of the parameters from the student. Consequently, the teacher prediction for each example can be regarded as an ensemble output of the current model version and its earlier versions. This ensembling enhances the quality of the predictions, leading to improved results when utilized as teacher predictions. Self-distillation has been explored in semi-supervised learning [39], self-supervised learning [32], [45], and HSI classification [46], [47]. In this article, we use predictions supervised by complete HSI patches for guidance in MSD to learn class-relevant knowledge.

B. Masked Image Modeling

MIM uses a partly masked image as input and performs a reconstruction of the uncorrupted image to extract knowledge from unlabeled data. With the success of masked language modeling, MIM has attracted increasing attention. Initial models such as iGPT [48] and iBERT [49] were designed to adapt the masked prediction task on image data. Subsequently, BEiT [50] demonstrated the power of MIM in extracting general and robust representations for vision transformer (ViT). Building upon this, MAE [51] and SimMIM [52] introduced an end-to-end method for training a masked autoencoder and achieved excellent performance. In summary, MIM created a pretext task on unlabeled image data by masking random parts of the input image and reconstructing the missing pixels. This forced the model to utilize the remaining context regions to infer the missing pixels. Due to the high masking ratio, mere pixel-level copying becomes insufficient for the pretext task. Consequently, the model is compeled to learn semantic representations from the unlabeled images to achieve reconstruction accuracy.

Unlike previous methods only based on transformer networks, ConvMAE [53] extended the MIM approach to convolutional neural networks. In addition, MIC [54] incorporated a spatially MSD strategy into UDA to enhance contextual relationship learning for semantic segmentation, but it yielded limited performance on classification. In the field of HSI classification, several advancements have been made with the application [55], [56], [57], [58], [59], [60], [61] of MIM, but there has been no research investigating the application of MIM to the cross-scene HSI classification task.

III. Proposed Method

The proposed MSDA method focuses on UDA, involving a source domain and a target domain with identical categories. In the source domain, n_s samples and their corresponding labels are denoted as $\boldsymbol{X}^s = \{\boldsymbol{x}_i^s\}_{i=1}^{n_s} \in \mathbb{R}^d$ and $\boldsymbol{Y}^s = \{y_i^s\}_{i=1}^{n_s} \in \{1, 2, \ldots, \mathcal{C}\}$, respectively, where d is the dimension of data, and $\mathcal{C}$ is the number of categories. In the target domain, n_t samples are represented as $\boldsymbol{X}^t = \{\boldsymbol{x}_i^t\}_{i=1}^{n_t} \in \mathbb{R}^d$. Note that, there are no target labels $\boldsymbol{Y}^t$ in the UDA case.

As illustrated in Fig. 1, the MSDA method consists of two primary components: an MSD module, denoted by green lines in the bottom part of Fig. 1, and a CSAT module, denoted by blue and orange lines in the top part of Fig. 1. The following parts first describe the backbone network architecture. Subsequently, we introduce the MSD module, which is designed to enhance the extraction of class-relevant knowledge within the

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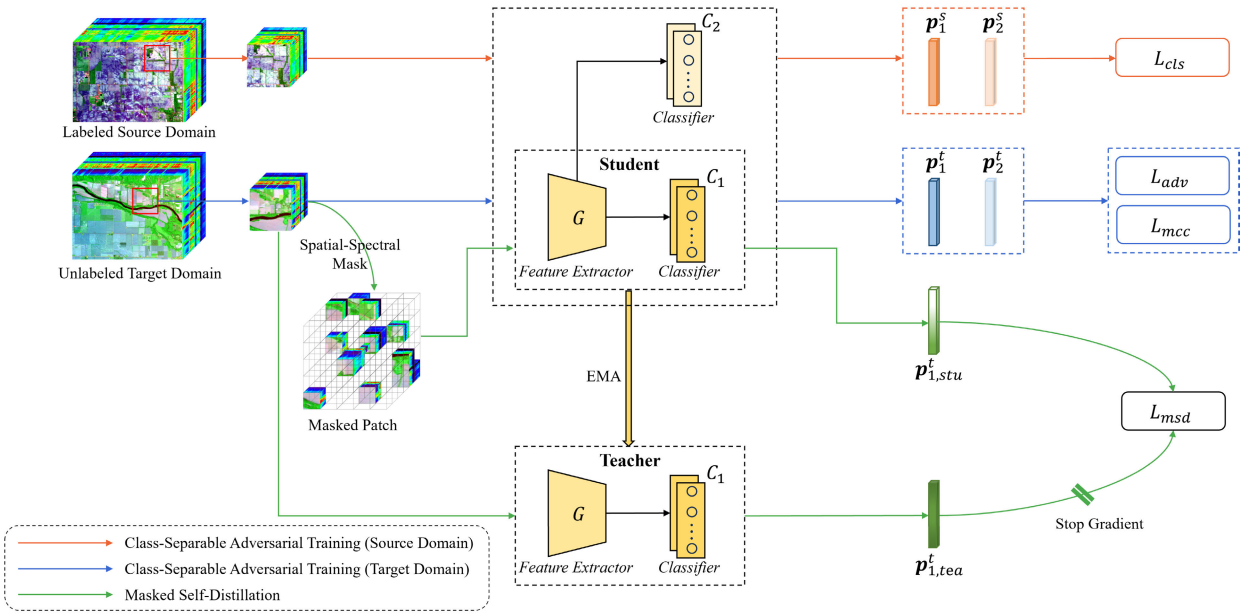


Fig. 1. Flowchart of MSDA. The complete source and target samples are passed through feature extraction and two classifiers, generating predictions that are used to calculate $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{adv}}$, and $\mathcal{L}_{\text{mcc}}$. The complete target samples are processed through the teacher network, and compared with the spatial–spectral masked target samples, which are processed through the student network, using $\mathcal{L}_{\text{msd}}$.

target samples. The CSAT scheme is then elaborated, which facilitates domain alignment while mitigating misclassification between ambiguous categories. Finally, the training pipeline is described.

A. Backbone Network

The proposed backbone network includes one feature extractor G and two classifiers C_1 and C_2, where the feature extractor G extracts spatial–spectral features and the two classifiers C_1 and C_2 make predictions using these features, as shown in the top part of Fig. 1. The detail architectures of the feature extractor and classifiers are shown in Figs. 2 and 3, respectively. Notably, instead of utilizing a single HSI pixel, we employ an HSI patch as the network input to extract spatial–spectral features, where the central pixel corresponds to the object to be classified.

For the feature extractor G, following the design of TAADA [30], we adopt a two-branch structure, which separately extracts spectral and spatial features through its two branches, and then fuses them. On the spectral branch, taking the Houston18 data as an example, the input sample with 7×7 patch size and 48 bands goes through three feature extraction blocks, each comprising 3-D convolution, batch normalization, and ReLU layers. Subsequently, the output is processed through an additional block to ensure that the dimensions of the spectral branch features align with those of the spatial branch features. The resulting features are then fed into the spectral attention module, which is employed to locate crucial spectral information within the sample. Specifically, the spectral attention calculates the importance weights of each spectral location, enabling the model to focus on more distinguishable spectral bands. By multiplying the spectral attention with the original spectral features, the model effectively weights the original features, amplifying the values of important regions while suppressing those of less important regions. On the spatial branch, the sample is also passed through three feature extraction blocks, which consist of 2-D convolution, batch normalization, and ReLU layers, and followed by a spatial attention module. This module enables the model to locate crucial spatial information within the sample and generate spatial features. Ultimately, the obtained spectral and spatial features are concatenated to simultaneously incorporate both spatial and spectral characteristics, resulting in spatial–spectral features.

The two classifier neural networks C_1 and C_2 possess the same structure, comprising a sequence of linear, batch normalization, ReLU, dropout, and additional linear layers. Following a softmax operation, they can produce the probability of a sample belonging to each class.

In the backbone networks, attention modules of the feature extractor G are beneficial for the model to focus on important regions within the samples, thereby extracting class-relevant information. In addition, biclassifier C_1 and C_2 can be used for adversarial training, which facilitates domain alignment.

B. Masked Self-Distillation

The MSD module only performs on the target samples, combining self-distillation with spatial–spectral masking, as shown in the bottom part of Fig. 1 with the green lines. First, an HSI patch is cut from the target domain to form a sample $\boldsymbol{x}^t \in \mathbb{R}^{\text{ps}\times\text{ps}\times\text{ch}}$, where ps and ch are the patch size and the channel number of the sample, respectively. Then, we apply a masking strategy to the sample $\boldsymbol{x}^t$ to obtain a masked sample $\boldsymbol{x}^m$. Subsequently, the masked sample $\boldsymbol{x}^m$ is fed into the student networks to produce a prediction $\boldsymbol{p}^t_{1,\text{stu}}$, while the unmasked sample $\boldsymbol{x}^t$ is input to the teacher networks and generate a prediction $\boldsymbol{p}^t_{1,\text{tea}}$. To ensure consistency between $\boldsymbol{p}^t_{1,\text{stu}}$ and $\boldsymbol{p}^t_{1,\text{tea}}$, the student networks must fully exploit the

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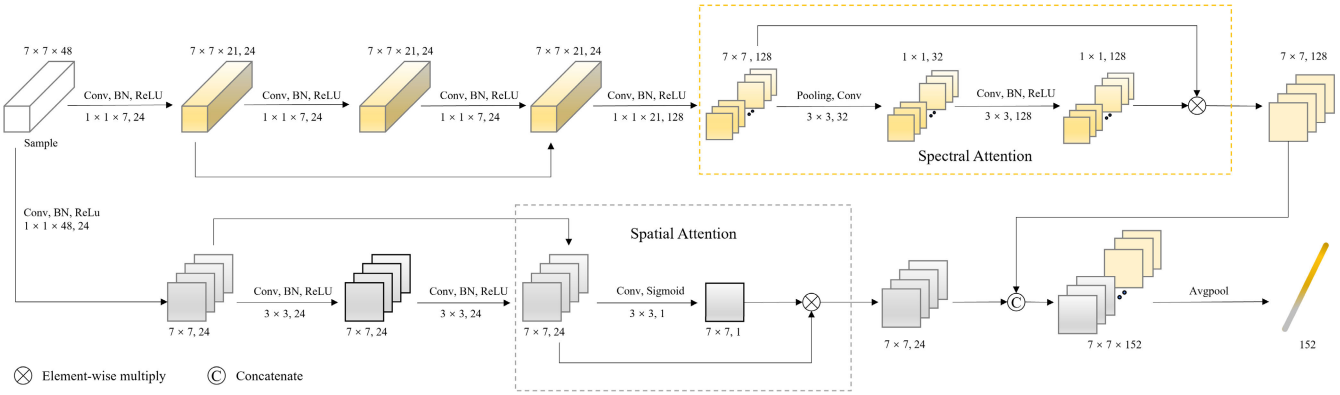


Fig. 2. Structure of feature extractor. The spatial and spectral features are acquired separately through the spatial branch (gray) and spectral branch (yellow). Subsequently, these features are concatenated to obtain spatial–spectral features.

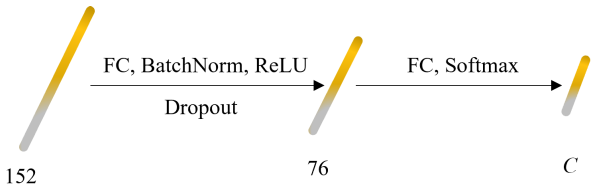


Fig. 3. Structure of classifier. FC represents the fully connected layer, and C represents the total number of classes. The features generated by the feature extractor go through two fully connected layers to obtain the probability of a sample belonging to each class.

information within the sample $\boldsymbol{x}^m$. This process is designed to enable class-relevant knowledge learning on target domains.

1) Spatial–Spectral Masking: In recent research, it has been shown that partial image information is sufficient to obtain intrinsic image characteristics [51], [52]. During the masking process of RGB images, a certain number of pixels are selected based on the masking ratio to cover all the information of the selected pixels. While RGB images only have three bands, HSI contains richer information in their numerous bands. Simply selecting a few pixels based on the masking ratio and masking all bands is not appropriate.

Given the characteristics of HSI, we make the following two improvements to the masking process.

1) Because the label of the central pixel in the sample is the label for the entire sample, the central pixel contains class-relevant information. Therefore, we do not mask the information of the central pixel during the masking process.
2) As HSI contains information from many bands and adjacent bands tend to be highly correlated, we employ a spatial–spectral masking strategy and mask HSI patch at snippet level.

Specifically, as illustrated in the masked patch of Fig. 1, we first divide the sample of size ps × ps × ch into several snippets of size $h \times h \times s$, and mask at the snippet level, where h represents the height and width of a snippet, and s represents the number of spectral bands associated with a snippet. We simultaneously retain all spectral band information of the central pixel.

For the masking process, we will first construct a mask tensor $\mathcal{M}$

$$\mathcal{M}_{\substack{ah+1:(a+1)h,\\ bh+1:(b+1)h,\\ es+1:(e+1)s}} = [v > r] \quad \text{with } v \sim \mathcal{U}(0,1) \tag{1}$$

where $[\cdot]$ denotes Iverson bracket, r denotes mask ratio, and $a, b \in [0, 1, \ldots, \lceil \text{ps}/h \rceil - 1]$, $e \in [0, 1, \ldots, \lceil \text{ch}/s \rceil - 1]$. Second, we set all bands of the central pixel in the mask to one to preserve the category information of samples. After obtaining the masking tensor, we multiply the samples with the corresponding elements of the masking tensor as

$$\boldsymbol{x}^m = \mathcal{M} \odot \boldsymbol{x}^t \tag{2}$$

where $\odot$ denotes element-wise multiplication. After the above processing, we can obtain the masked sample.

2) Self-Distillation: The masked sample can only provide limited information by its remaining regions, making the prediction more difficult. To encourage the networks to leverage the remaining information for accurate label prediction, we employ self-distillation learning. In this approach, the original network acts as the student model. Specifically, we form the student networks using a combination of G and C_1, and create teacher networks G_{tea} and $C_{1,\text{tea}}$ with the same structure by deep copying θ_G and θ_{C_1}.

During training, a stop-gradient operator is applied on the teacher networks to propagate gradients only through the student networks, as shown in Fig. 1. The parameters of the teacher networks are updated using EMA, with the following expression:

$$\phi_{t+1} \leftarrow \alpha\phi_t + (1-\alpha)\theta_t \tag{3}$$

where ϕ represents parameters of the teacher networks, θ represents parameters of the student networks, and t denotes training step. Therefore, the EMA teacher realizes a temporal ensemble of previous student models, which increases the robustness and accuracy of its prediction [32].

For the target sample, we pass the unmasked sample $\boldsymbol{x}^t$ through the teacher networks, and obtain the probability of $\boldsymbol{x}^t$ belonging to each class

$$g_\phi(\boldsymbol{x}^t) = C_{1,\text{tea}}(G_{\text{tea}}(\boldsymbol{x}^t)). \tag{4}$$

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Since the target domain has unlabeled data, we use this method to generate the pseudo-labels

$$\hat{y}(\boldsymbol{x}^t) = \left[c = \arg\max_{c'} g_\phi(\boldsymbol{x}^t)_{c'} \right]. \tag{5}$$

Then, we predict the class of the masked sample by the student networks. We aim for the predictions from the two networks to be as consistent as possible. The difference between the two predictions is calculated using MSD loss as follows:

$$\mathcal{L}_{\text{msd}}(\boldsymbol{x}^t, \boldsymbol{x}^m) = \frac{q^t}{2} \mathcal{L}_{\text{ce}}(C_1(G(\boldsymbol{x}^m)), \hat{y}(\boldsymbol{x}^t)) \tag{6}$$

where C_1 and G denote the student network classifier and student network feature extractor, respectively. $\mathcal{L}_{\text{ce}}(\cdot)$ represents cross-entropy loss function. q^t denotes the quality weight, which can be calculated using maximum softmax probability as follows:

$$q^t = \left[c = \max_{c'} g_\phi(\boldsymbol{x}^t)_{c'} \right]. \tag{7}$$

Through the use of the MSD loss, the student networks are encouraged to aim for a better prediction obtained with the teacher networks and achieve more accurate classification of samples with only partial data. Notably, the teacher networks also improve since it is built directly from the student weights. This iterative refinement progressively improves the student network performance, ultimately leading to enhanced classification accuracy.

C. Class-Separable Adversarial Training

To achieve domain alignment, we leverage a feature extractor G and two classifiers C_1 and C_2 in an adversarial training manner, as shown in the top part of Fig. 1 with blue and orange lines. In this process, the class confusion problem can arise due to the presence of unlabeled ambiguous categories in the target domain. To mitigate this issue and extract class-separable features, we introduce an MCC loss that prevents misclassification of target samples during marginal distribution alignment.

1) Minimize Class Confusion: It is expected that the sample classification results are as correct as possible after passing through the backbone networks. For source samples, the ground-truth labels are used as supervision, which allows the model to classify the samples correctly. Yet, for the target domain samples, there are no truth labels, so it is difficult to guide them correctly. In order to solve this problem, we adopt the MCC loss to improve the separability of features, and we hope to find more valuable samples in the target domain and more discriminative features through MCC. In Section III-C1a–III-C1d, we will introduce the implementation process of the MCC loss carefully. The probability $p_i^t = C_i(G(\boldsymbol{x}^t)) \in \mathbb{R}^{B*|\mathcal{C}|}, i \in \{1, 2\}$ is generated by the two classifiers of the backbone network with B being the batch size. For clarity, we omit domain superscripts and classifier subscripts in the following.

a) Probability rescaling: In previous research, deep neural networks often produce excessively confident predictions, which can impede their ability to effectively address class confusion. Consequently, we employ temperature rescaling as in (8), a straightforward yet powerful technique, to mitigate the adverse impact of overconfident predictions

$$z_{ij} = \frac{\exp(p_{ij}/T)}{\sum_{j=1}^{|\mathcal{C}|} \exp(p_{ij}/T)} \tag{8}$$

where p_{ij} denotes the probability that the ith sample belonging to the jth class, and T is a hyperparameter representing the temperature.

b) Class correlation: As z_{ij} reveals the relationship between the ith instance and the jth class, we define the class correlation between two classes j and j' as

$$R_{jj'} = \boldsymbol{z}_{\cdot j}^{\top} \boldsymbol{z}_{\cdot j'} \tag{9}$$

which quantifies the likelihood of the classifier simultaneously categorizing the B examples into both the jth and j'th classes.

c) Uncertain weight: We note that examples are not equally important for quantifying class confusion. When the prediction is closer to a uniform distribution, lacking obvious peaks (significantly larger probabilities for some classes), we consider the classifier ignorant about this example. In contrast, when the prediction exhibits multiple peaks, it signifies the classifier's ambiguity between several confusing classes. Evidently, examples rendering the classifier ambiguous across classes are more suitable to characterize class confusion.

We employ the entropy function to measure the level of uncertainty associated with a sample. The expression is as follows:

$$H(\boldsymbol{z}_{i\cdot}) = -\sum_{j=1}^{|\mathcal{C}|} z_{ij} \log z_{ij}. \tag{10}$$

On the other hand, we aim to place greater emphasis on distributions with higher accuracy, so we apply a softmax transformation. The transformed weight expression is

$$W_{ii} = \frac{B(1 + \exp(-H(\boldsymbol{z}_{i\cdot})))}{\sum_{i'=1}^{B}(1 + \exp(-H(\boldsymbol{z}_{i'\cdot})))} \tag{11}$$

which is an element of the diagonal matrix $\boldsymbol{W}$.

d) MCC loss: With this weight mechanism, the class confusion can be rewritten from (9) as

$$\Delta_{jj'} = \boldsymbol{z}_{\cdot j}^{\top} \boldsymbol{W} \boldsymbol{z}_{\cdot j'}. \tag{12}$$

Nonetheless, with a large number of classes, a severe class imbalance arises within each batch. To address this, we employ a category normalization technique extensively utilized in Random Walk algorithms, and the specific expressions are as follows:

$$\widetilde{\Delta}_{jj'} = \frac{\Delta_{jj'}}{\sum_{j''=1}^{|\mathcal{C}|} \Delta_{jj''}}. \tag{13}$$

Built upon the previous theories, our method can be established for MCC. The off-diagonal elements need to

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be minimized for accurate predictions for each class. The objective is to minimize the class confusion loss defined as

$$\mathcal{L}_{\text{mcc}}(\boldsymbol{x}^t) = \frac{1}{|\mathcal{C}|}\sum_{j=1}^{|\mathcal{C}|}\sum_{j'\neq j}^{|\mathcal{C}|}\left|\widetilde{\Delta}_{jj'}\right|. \tag{14}$$

2) Distribution Alignment: In UDA, since label information is absent, the objective is to align the marginal distributions between the source and target domains as closely as possible. Here, we employ a three-step adversarial training approach, similar to that used in [35]. In this process, for more accurate classification, we add the $\mathcal{L}_{\text{mcc}}$ and $\mathcal{L}_{\text{msd}}$ to the domain adaptation process.

At first, we utilize the source samples $\boldsymbol{x}^s$, source labels y^s and target samples $\boldsymbol{x}^t$ to train the entire neural networks, and the loss expression is as follows:

$$\min_{\theta_g,\theta_{c_1},\theta_{c_2}} \mathcal{L}_{\text{cls}}\left(\boldsymbol{x}^s, y^s\right) + \mathcal{L}_{\text{msd}}\left(\boldsymbol{x}^t, \boldsymbol{x}^m\right) + \mathcal{L}_{\text{mcc}}\left(\boldsymbol{x}^t\right) \tag{15}$$

where $\mathcal{L}_{\text{cls}}$ stands for classification loss, which is defined as follows:

$$\mathcal{L}_{\text{cls}}\left(\boldsymbol{x}_i^s, y_i^s\right) = \frac{1}{2n_s}\sum_{i=1}^{n_s}\sum_{j=1}^{2}\mathcal{L}_{\text{ce}}\left(C_j\left(G\left(\boldsymbol{x}_i^s\right)\right), y_i^s\right) \tag{16}$$

where $C_j(\cdot)$ is the output of the jth classifier C_j, and y_i^s represents the ground-truth label of the ith sample from the source domain. In this step, by minimizing $\mathcal{L}_{\text{cls}}$ using outputs $\boldsymbol{p}_1^s$ and $\boldsymbol{p}_2^s$ of the two classifiers, the networks are able to accurately classify the source samples. Since both $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{msd}}$ are cross-entropy losses, we use them together.

Second, we fix the feature extractor G and update the parameters θ_{C_1} and θ_{C_2} which belong to the classifier C_1 and C_2. Both source and target domain data are given to the G and two sets of predictions $\boldsymbol{p}_1^t$ and $\boldsymbol{p}_2^t$ are generated from the classifiers. The objective is to preserve source domain classification accuracy while maximizing the discrepancy between the classifier C_1 and classifier C_2. During this process, the loss is expressed as

$$\min_{\theta_{c_1},\theta_{c_2}} \mathcal{L}_{\text{cls}}\left(\boldsymbol{x}^s, y^s\right) + \mathcal{L}_{\text{msd}}(\boldsymbol{x}^t, \boldsymbol{x}^m) - \mathcal{L}_{\text{adv}}(\boldsymbol{x}^t) + \mathcal{L}_{\text{mcc}}(\boldsymbol{x}^t) \tag{17}$$

where $\mathcal{L}_{\text{adv}}$ denotes the adversarial loss between the two classifiers, defined as

$$\mathcal{L}_{\text{adv}}\left(\boldsymbol{x}_i^t\right) = \frac{1}{n_t}\sum_{i=1}^{n_t}\left\|C_1\left(G\left(\boldsymbol{x}_i^t\right)\right) - C_2\left(G\left(\boldsymbol{x}_i^t\right)\right)\right\|_1 \tag{18}$$

where n_t is the total number of training samples in the target domain and $\|\cdot\|_1$ represents the Manhattan norm.

Finally, the two classifiers C_1 and C_2 are fixed, and the feature extractor G is optimized to align the results $\boldsymbol{p}_1^t$ and $\boldsymbol{p}_2^t$ of these two classifiers, ensuring consistency in their outputs. This step aims to generate domain-invariant features. The detailed definition of this alignment process is as follows:

$$\min_{\theta_g} \mathcal{L}_{\text{adv}}(\boldsymbol{x}^t) + \mathcal{L}_{\text{mcc}}(\boldsymbol{x}^t). \tag{19}$$

During training, these three steps are iteratively repeated. Its purpose is to enable the feature extractor to generate domain-invariant spectral–spatial features while the classifiers can accurately classify the source and target samples.

D. Training Pipeline

In our method, MSDA achieves the reduction of domain discrepancies and the extraction of highly discriminative features from the target domain data by integrating CSAT and MSD strategies. Specifically, in the first stage of MSDA, the feature extractor and the two classifiers are updated using the classification loss, MSD loss, and MCC loss, as described in (15). In the second stage, the two classifiers are updated using the classification loss, adversarial loss, MSD loss, and MCC loss, as described in (17). In the third stage, the feature extractor is updated using the adversarial loss and MCC loss, as described in (19). The final pseudocode is detailed in Algorithm 1.

Algorithm 1 Training Pipeline of MSDA for UDA

Input: Source domain samples $\boldsymbol{x}^s \in \boldsymbol{X}^s$ and labels $y^s \in \boldsymbol{Y}^s$, target domain samples $\boldsymbol{x}^t \in \boldsymbol{X}^t$, hyperparameter ps, h, s, r and T, the total epoch number eps.

Output: The parameters θ_G of feature extractor G, and θ_{C_1} of classifier C_1.

```
Randomly initialize parameters θ_{C1}, θ_{C2}, and θ_G of classifiers C1, C2, and feature extractor G.
Initialize parameters φ_G, φ_{C1} of teacher networks G_tea, C_{1,tea} by copying parameters θ_G, θ_{C1}.
for epoch in {1, 2, ..., eps} do
    Updating φ_G and φ_{C1} by Eq. (3).
    Generate x^m using x^t by Eq. (2).
    Update θ_G, θ_{C1}, θ_{C2} using x^s, y^s, x^t and x^m through Eq. (15).
    Update θ_{C1}, θ_{C2} using x^s, y^s, x^t and x^m through Eq. (17).
    Update θ_G using x^t by Eq. (19).
end for
```

IV. Experiment

A. Data Description

To evaluate the effectiveness of the proposed MSDA method, we conducted five experiments on eight datasets: Houston13, Houston18, University of Pavia, Pavia Center, Dioni, Loukia, NC12, and NC13.

1) H13Train→H13Test: The Houston13 (H13) dataset was acquired in June 2012 by the Compact Airborne Spectrographic Imager (CASI) 1500 sensor over the University of Houston campus and its surrounding urban area. This dataset comprises 144 spectral channels with dimensions of 349 × 1905 pixels, covering a wavelength range from 364 to 1046 nm, and boasts a high spatial resolution of 2.5 m/pixel. The ground information is divided into different subsets, which are used separately for training and testing purposes, collectively encompassing 15 mutually exclusive urban categories with a total of 15 029 annotated samples. Due to cloud cover, which leads to different illumination conditions, there exists a domain shift between the training and testing sets. The

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TABLE I
LAND COVER CLASSES AND NUMBER OF SAMPLES IN H13TRAIN→H13TEST TASK

No.	Class	H13Train (Source)	H13Test (Target)
1	Grass healthy	198	1053
2	Grass stressed	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential building	196	1072
8	Non-residential building	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
	Total	2832	12197

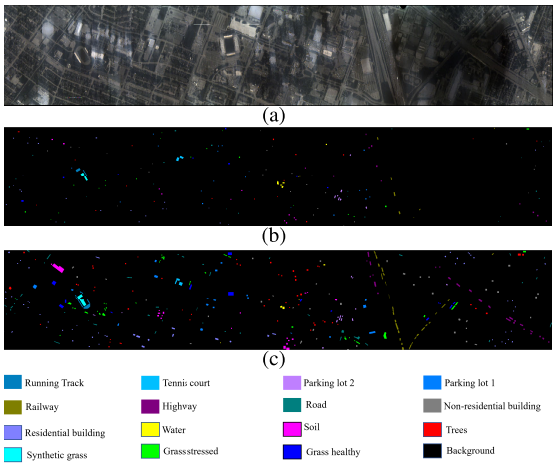


Fig. 4. H13Train→H13Test image. (a) False-color image of Houston13. (b) Ground-truth map of H13Train. (c) Ground-truth map of H13Test.

training set (H13Train) serves as the source domain, while the testing set (H13Test) is treated as the target domain. The source domain contains 2832 samples, and the target domain comprises 12 197 samples. Table I shows the class names and the number of samples in the source and target domains in the H13Train→H13Test task. Fig. 4 displays the false-color images and ground-truth maps.

2) H13→H18: The Houston18 (H18) dataset was acquired in February 2017, containing 209 × 955 pixels with 48 spectral bands, with a spatial resolution of 1 m. H13 and H18 have the same wavelength range. There are significant temporal and spatial differences between the H13 and H18 datasets, resulting in domain shifts. Specifically, in this task, on the H13 dataset, we chose 209 × 955 overlapping areas and 48 spectral bands, which correspond to Houston18. Table II shows the

TABLE II
LAND COVER CLASSES AND NUMBER OF SAMPLES IN H13→H18 TASK

No.	Class	H13 (Source)	H18 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

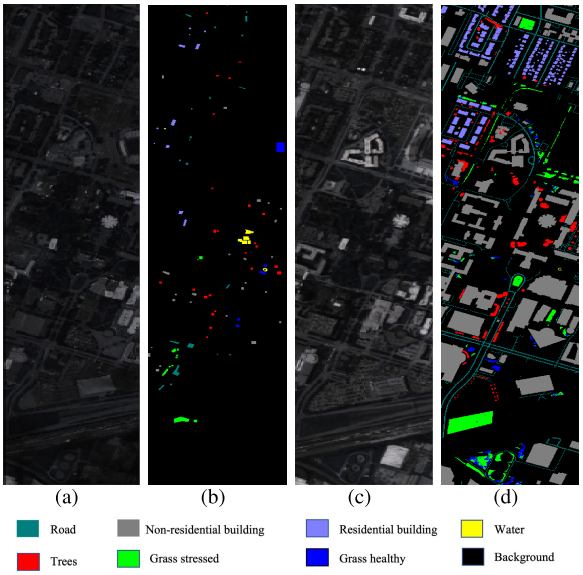


Fig. 5. H13→H18 image. (a) False-color image of H13. (b) Ground-truth map of H13. (c) False-color image of H18. (d) Ground-truth map of H18.

class names and the number of samples in the source and target domains in the H13→H18 task. Fig. 5 displays the false-color images and ground-truth maps.

3) UP→PC: In this task, we include two datasets: the University of Pavia (UP, source domain) and Pavia Centre (PC, target domain). Due to differences in their acquisition areas, there are domain discrepancies between them. Both datasets were collected by ROSIS sensors in Pavia in northern Italy. The UP dataset has 610 × 610 pixels and 103 bands, and the PC dataset has 1096 × 1096 pixels and 102 bands. Specifically, because some samples in these two domains have no information, they can be discarded before analysis. Therefore, UP contains 610 × 315 pixels, and PC contains 1096 × 715 pixels. For UP, the last band was removed to ensure the same number of spectral bands as PC. In particular, after removing the last band of the UP, the spectral wavelength range of both datasets is 430–834 nm. They all have seven same categories. Table III shows the class names and the number of samples in the source and target domains. Fig. 6 displays the false-color images and ground-truth maps.

4) Dioni→LouKia: In this task, Dioni (source domain) and Loukia (target domain) [62] were developed within the

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TABLE III
LAND COVER CLASSES AND NUMBER OF SAMPLES IN UP→PC TASK

No.	Class	UP (Source)	PC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

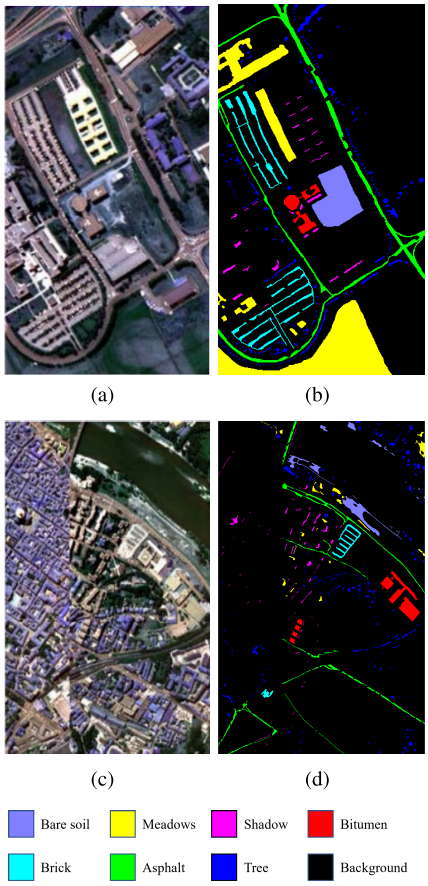


Fig. 6. UP→PC image. (a) False-color image of UP. (b) Ground-truth map of UP. (c) False-color image of PC. (d) Ground-truth map of PC.

TABLE IV
LAND COVER CLASSES AND NUMBER OF SAMPLES IN DIONI→LOUKIA TASK

No.	Class	Dioni (Source)	LouKia (Target)
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sclerophyllous vegetation	5035	2996
8	Sparce sclerophyllous vegetation	6374	2361
9	Sparcely vegetated areas	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
	Total	20024	10317

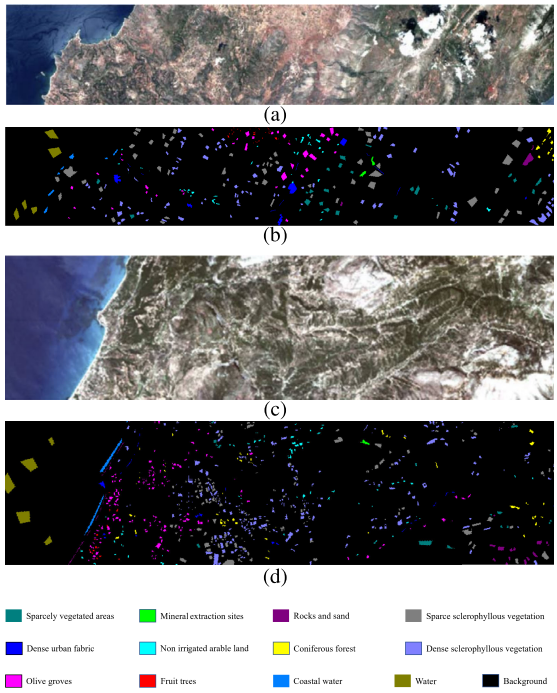


Fig. 7. Dioni→LouKia image. (a) False-color image of Dioni. (b) Ground-truth map of Dioni. (c) False-color image of LouKia. (d) Ground-truth map of LouKia.

framework of the Scientific Initiatives of the International Society for Photogrammetry and Remote Sensing (ISPRS). The satellite hyperspectral imagery was acquired by the Hyperion sensor aboard the Earth Observing 1 (EO-1) satellite mission operated by the United States Geological Survey (USGS). Hyperion, onboard the Earth Observing-1 satellite, was the first spaceborne hyperspectral instrument capable of acquiring spectra spanning the visible near-infrared (VNIR, 400–1000 nm) and shortwave infrared (SWIR, 900–2500 nm) regions. After data preprocessing, each hyperion hyperspectral datacube has 176 remaining bands. The Dioni dataset has 250 × 1376 pixels, the Loukia dataset has 250 × 945 pixels, and the source and target domains contain 12 consistent land cover classes. Table IV shows the class names and the number of samples in the source and target domains in the Dioni→LouKia task. Fig. 7 displays the false-color images and ground-truth maps.

5) NC12→NC13: The NC12 and NC13 datasets [63] were acquired by a 12-mm focal length headwall nano-hyperspec imaging sensor equipped on DJI Matrice 600 (DJI M600) unmanned aerial vehicle (UAV) platform in the Yellow River Delta National Nature Reserve in Shandong Province, China, and the sensor records 270 bands in a wavelength range

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TABLE V
LAND COVER CLASSES AND NUMBER OF SAMPLES IN NC12→NC13 TASK

No.	Class	NC12 (Source)	NC13 (Target)
1	Suaeda glauca	14117	14225
2	Reed	15603	56922
3	Water	36291	63007
4	Tamarix	9257	1954
5	Moist soil	68497	4910
6	Dry soil	120073	4427
7	Standard reflectance cloth	75	49
	Total	263913	145494

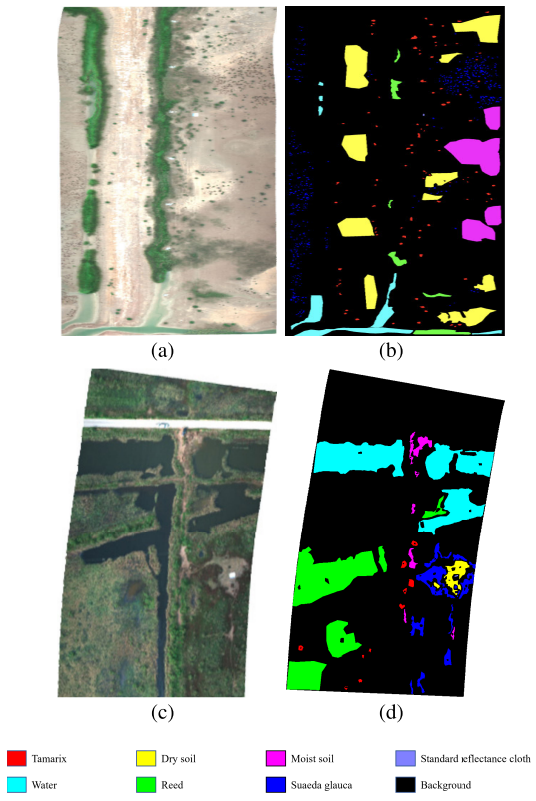


Fig. 8. NC12→NC13 image. (a) False-color image of NC12. (b) Ground-truth map of NC12. (c) False-color image of NC13. (d) Ground-truth map of NC13.

from 400 to 1000 nm. The NC12 dataset was acquired on September 23, 2020, from 12:27 to 13:10, under clear and cloudless weather conditions. The spatial resolution of the NC12 dataset is about 0.182 m because the flying height of the UAV is 300 m. There are 12 types of land covers. In addition, the NC13 dataset was acquired from 14:47 to 15:20 on September 24, 2020, with a cloudy day. The flying height of the UAV is 300 m, and the spatial resolution of the NC13 dataset is also about 0.182 m. These scenes encompass seven consistent land cover classes. Table V shows the class names and number of samples in the source and target domains in the NC12→NC13 task. Fig. 8 displays the false-color images and ground-truth maps.

B. Experimental Setup

The proposed MSDA method is compared with seven state-of-the-art approaches, including DAN [23], DANN [26], ED-DMM-UDA [27], CDA [28], CLDA [29], TAADA [30], and SCLUDA [25], to demonstrate its effectiveness. In the experiments, the H13Train→H13Test task uses all source samples for training since it has a fixed data division. In all the other tasks, we take 180 samples from each category in the source domain as the training data.

For DAN, ResNet50 is used as the feature extractor, with three adaptive layers using MK-MMD metric, and the number of different Gaussian Kernels is 5. The network is trained for 1000 iterations with a batch size of 32. The optimizer is SGD with a momentum of 0.9 and weight decay of 0.0005. For DANN, ResNet18 architecture is adopted as the feature extractor, and a three-fully connected-layer classifier is used to discriminate between source and target. The batch size, learning rate, and number of epochs are 32, 0.01, and 1000, respectively. For ED-DMM-UDA, the source epoch is set to 1000, and the target epoch is set to 20. For CDA, the batch size is set to 32 and the learning rate is set to 0.001, 0.05, 0.05, 0.05, and 0.05 in five tasks, and the ratio of $\lambda 1$ and $\lambda 2$ is set as 10:1, 1:1, 1:1, 1:1, 1:1, and 1:1. For SCLUDA, the settings for H13→H18 follow the original paper, and α, β, and γ is set to 0.01, 0.01, and 0.03 in UP→PC, Dioni→LouKia, NC12→NC13, and H13Train→H13Test. In all tasks, TRAIN_NUM is 100, ITER_NUM is 40. For TAADA, all experimental settings are consistent with the original paper. For CLDA, the configuration of H13→H18 and UP→PC are the same as in the origin paper, α, β, and γ is 0.01, 0.1, and 0.03 in Dioni→LouKia, NC12→NC13, and H13Train→H13Test. Other parameters of all the methods are consistent with the original paper.

In our MSDA experiments, the learning rate is 0.1, and the batch size is 32. After training for 50 epochs, we evaluate the classification accuracy of the model on the target domain. It is worth noting that the datasets in H13Train→H13Test, UP→PC, and Dioni→LouKia obey the standard normal distribution $N(0, 1)$, and the datasets in H13→H18 and NC12→NC13 are not processed. The mask ratio r is selected between 0.1 and 0.9, h is chosen between 1 and $\lceil ps/2 \rceil$, s is between 1 and 15, and the temperature T for $\mathcal{L}_{mcc}$ is selected between 1 and 10. Within the H13Train→H13Test task, the HSI patch size is established at $9 \times 9 \times 144$, with snippet dimensions of $3 \times 3 \times 15$, r is set to 0.1, and T is adjusted to 1. On the H13→H18 task, the size of the sample is $7 \times 7 \times 48$, with snippet dimensions of $2 \times 2 \times 6$, the masking ratio r is established to 0.1, and the hyperparameter T is configured to 9. For the UP→PC task, the samples are sized at $9 \times 9 \times 102$, the snippets are dimensioned at $4 \times 4 \times 9$, r is adjusted to 0.7, and T is also set to 9. In the context of the Dioni→LouKia task, the sample dimensions are $3 \times 3 \times 176$, with snippets measured at $2 \times 2 \times 13$, the masking ratio r is defined as 0.7, and T is determined to be 7. Finally, within the NC12→NC13 task, the sample size is established at $3 \times 3 \times 270$, with snippet dimensions of $3 \times 3 \times 5$, r is set to 0.9, and T is adjusted to 6.

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TABLE VI
CLASSIFICATION RESULTS* ON H13TRAIN→H13TEST TASK

Class	DAN [23]	DANN [26]	ED-DMM-UDA [27]	CDA [28]	CLDA [29]	TAADA [30]	SCLUDA [25]	MSDA
1	81.29	79.58	82.43	81.67	81.86	83.1	**83.1**	82.81
2	80.64	83.65	85.06	84.02	85.15	85.15	85.15	**85.15**
3	63.96	32.08	100	99.6	**100**	99.8	99.41	99.6
4	86.17	76.42	89.58	89.87	92.8	92.52	89.77	**93.28**
5	96.31	98.11	**100**	97.25	99.91	**100**	**100**	**100**
6	75.52	74.13	**95.8**	94.41	**95.8**	**95.8**	**95.8**	**95.8**
7	80.32	80.78	79.1	72.11	83.49	82.65	**88.34**	84.98
8	50.33	58.21	43.97	45.68	50.9	76.64	75.21	**78.73**
9	75.45	79.89	**89.05**	75.83	84.14	84.51	81.59	88.29
10	46.14	46.81	28.86	44.4	68.53	68.24	52.03	**99.81**
11	58.35	57.12	69.07	73.15	79.03	84.54	**94.88**	86.53
12	81.46	82.61	95.29	78.19	65.61	98.75	95.87	**98.75**
13	84.56	83.16	0.7	71.93	91.93	90.18	**92.98**	**92.98**
14	80.16	85.02	**100**	98.38	**100**	**100**	**100**	**100**
15	84.78	60.47	**100**	98.31	**100**	**100**	**100**	**100**
OA(%)	74.14	72.51	77.17	76.93	81.78	87.28	86.49	**90.95**
AA(%)	75.03	71.87	77.26	80.32	85.28	89.46	88.94	**92.45**
$\kappa \times 100$	72.12	70.24	75.4	75.11	80.4	86.29	85.4	**90.19**

* In the case of fixed split datasets, where all source domain samples are used to train the model in a single pass, calculating the standard deviation of the results is unnecessary.

TABLE VII
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON H13→H18 TASK

Class	DAN [23]	DANN [26]	ED-DMM-UDA [27]	CDA [28]	CLDA [29]	TAADA [30]	SCLUDA [25]	MSDA
1	89.08	79.71	68.14	**95.08**	63.03	94.29	81.32	71.60
2	56.09	73.26	**85.45**	51.96	84.85	53.03	71.58	67.01
3	**79.55**	70.91	57.86	62.52	72.18	53.13	62.67	64.64
4	89.55	**98.18**	79.55	95.45	94.55	97.27	90.00	78.64
5	69.04	69.61	**97.77**	70.35	93.59	80.42	85.11	89.72
6	50.61	55.72	56.62	**82.74**	62.16	76.61	80.53	82.56
7	66.66	60.81	75.97	26.29	**79.26**	75.64	49.13	75.91
OA(%)	57.39 ±1.19	60.75 ±5.03	66.09 ±1.83	71.18 ±1.31	70.02 ±2.94	73.95 ±2.65	75.50 ±1.77	**79.84 ±1.27**
AA(%)	71.51 ±1.09	72.6 ±2.05	74.48 ±1.66	69.20 ±3.16	**78.53 ±2.06**	75.77 ±2.31	74.32 ±4.93	75.72 ±2.47
$\kappa \times 100$	43.25 ±1.16	45.85 ±4.45	53.10 ±2.18	51.85 ±3.23	57.44 ±3.11	60.29 ±3.35	61.00 ±3.38	**68.26 ±1.66**

TABLE VIII
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON UP→PC TASK

Class	DAN [23]	DANN [26]	ED-DMM-UDA [27]	CDA [28]	CLDA [29]	TAADA [30]	SCLUDA [25]	MSDA
1	72.79	84.49	**98.15**	92.02	96.58	91.50	95.97	96.22
2	79.37	90.51	94.12	77.63	**99.68**	98.72	98.44	98.89
3	87.50	75.86	64.02	86.90	78.07	57.07	**98.03**	85.23
4	67.76	71.28	0.72	78.17	80.07	69.94	**85.7**	84.75
5	98.75	99.58	96.61	99.94	**99.99**	97.50	98.42	99.93
6	80.22	69.52	68.29	61.81	86.96	92.94	95.03	**96.42**
7	58.37	62.67	64.79	71.60	90.59	**95.46**	78.13	95.00
OA(%)	74.47 ±2.07	79.14 ±2.68	68.80 ±2.58	80.51 ±1.02	91.48 ±2.26	88.07 ±5.97	91.91 ±0.44	**94.06 ±0.26**
AA(%)	77.82 ±1.90	79.13 ±1.96	69.53 ±2.41	81.15 ±1.81	90.28 ±1.96	86.16 ±6.71	92.82 ±0.60	**93.78 ±0.66**
$\kappa \times 100$	69.79 ±2.43	75.02 ±3.12	62.47 ±3.16	76.68 ±1.22	89.71 ±2.72	85.72 ±7.04	92.36 ±0.40	**92.86 ±0.31**

In the test phase, the MSD module is removed. The complete target samples are sent into the student networks, and their final prediction labels are obtained based on the outputs of classifier C_1, while the outputs of C_2 are discarded.

To evaluate the experimental results, we mainly use three metrics: OA, AA, and κ.

C. Experimental Results

Tables VI–X show OA, AA, and κ in different comparison methods for five different tasks, as well as the classification accuracy of each class. Except for H13Train→H13Test, all experiments were performed ten times to eliminate the effect of random sampling, and the mean of them was taken as the

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TABLE IX
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON DIONI→LOUKIA TASK

Class	DAN [23]	DANN [26]	ED-DMM-UDA [27]	CDA [28]	CLDA [29]	TAADA [30]	SCLUDA [25]	MSDA
1	33.98	19.81	**69.81**	30.05	26.70	45.68	36.94	27.28
2	**100**	96.30	91.48	85.93	97.59	**100**	71.85	94.81
3	57.09	31.48	59.65	77.84	61.15	76.95	56.83	**87.21**
4	74.81	69.75	40.51	28.35	77.09	65.44	**96.71**	56.33
5	35.98	19.79	47.53	58.84	45.74	71.62	12.97	**77.94**
6	31.35	26.28	12.61	42.77	49.64	5.73	54.83	2.13
7	49.78	43.05	78.19	50.02	43.16	41.64	**78.57**	51.57
8	31.79	30.47	49.25	22.64	28.52	50.7	27.69	**63.08**
9	61.88	31.68	64.26	23.31	28.37	26.74	**85.16**	45.86
10	54.64	31.37	54.97	**63.75**	51.77	20.68	50.42	35.74
11	99.25	89.39	70.00	60.02	**100**	**100**	**100**	**100**
12	76.94	95.11	68.08	99.98	**100**	**100**	**100**	**100**
OA(%)	52.34 ±1.51	43.96 ±4.49	61.37 ±9.04	48.14 ±0.06	51.11 ±4.56	56.25 ±2.96	60.08 ±2.47	**63.86 ±1.01**
AA(%)	59.96 ±1.58	48.71 ±3.49	58.86 ±3.40	53.63 ±4.56	59.14 ±5.09	58.77 ±3.23	**64.33 ±5.10**	61.83 ±1.58
$\kappa \times 100$	43.23 ±2.28	34.91 ±4.48	54.33 ±3.40	40.89 ±6.57	44.15 ±4.92	48.67 ±3.12	53.65 ±2.69	**56.95 ±1.09**

TABLE X
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON NC12→NC13 TASK

Class	DAN [23]	DANN [26]	ED-DMM-UDA [27]	CDA [28]	CLDA [29]	TAADA [30]	SCLUDA [25]	MSDA
1	94.23	37.77	68.40	30.53	88.08	99.14	61.21	**99.43**
2	68.09	77.13	46.96	69.05	77.53	79.35	74.53	**97.93**
3	16.89	20.20	70.35	**99.99**	96.42	**99.99**	90.77	**99.99**
4	82.50	56.55	31.02	45.71	64.01	**87.30**	47.26	32.31
5	49.00	31.53	2.75	48.35	0.09	**66.61**	13.65	11.27
6	15.75	7.99	25.20	0.00	88.57	0.01	**44.07**	16.93
7	91.43	68.37	75.71	**100**	83.64	98.16	**100**	**100**
OA(%)	46.44 ±5.05	44.71 ±18.80	56.83 ±25.39	75.58 ±2.85	83.64 ±7.33	87.49 ±5.54	76.92 ±13.46	**92.70 ±0.48**
AA(%)	59.7 ±3.07	42.79 ±10.09	45.77 ±10.43	56.23 ±2.13	67.92 ±6.14	**75.79 ±2.14**	61.64 ±6.67	65.41 ±3.22
$\kappa \times 100$	35.51 ±4.34	29.43 ±22.74	43.78 ±26.09	63.56 ±3.78	76.48 ±9.37	81.73 ±7.13	67.49 ±16.95	**88.68 ±0.69**

final classification result. We can see the following from the results.

For the H13Train→H13Test task. Table VI presents the experimental results of different compared algorithms on the H13Train→H13Test task. From the classification results, it can be observed that the accuracy of DAN, DANN, ED-DMM-UDA, and CDA are all around 75%, CLDA achieves an accuracy of 81.78%, TAADA and SCLUDA reach approximately 87%, and the MSDA method achieves the best performance with an accuracy of 90.95% and achieves better results in terms of the AA and the κ. And it is able to correctly target the real category of the domain under cloud cover. Fig. 9 displays the classification map of different compared algorithms. Despite cloud coverage in the target domain, we are still able to distinguish between Highway and Nonresidual building areas. In addition, our classification map is smoother.

Table VII presents the experimental results of different compared algorithms on the H13→H18 task. In terms of OA, DAN achieves an accuracy of 57.39%. DANN and ED-DMM-UDA gradually improved the accuracy of this task. CDA and CLDA increase the target domain accuracy to around 70%. TAADA and SCLUDA reach 73.95% and 75.50%, respectively, while our method achieves 79.84% accuracy on the target domain. For AA, CLDA performs the best among the compared methods. For the κ, our method obtains the best result of 69.26%. Fig. 10 displays the classification map of different compared algorithms. Our classification map is smoother, and we achieve more accurate differentiation in the tree regions.

For the UP→PC task, Table VIII details the comparative experimental results of the various algorithms. The source and target domains for these experiments were the UP and PC scenes. In terms of OA, ED-DMM-UDA records a notable 74.47%. Both DANN and CDA achieve impressive accuracies, hovering around 80%. Remarkably, CLDA, TAADA, and SCLUDA show significant advancements over the preceding methods, with each registering accuracies in the vicinity of 90%. Our method outperforms these, attaining a remarkable 94.06% accuracy on PC. Furthermore, our method secures an AA of 93.78%. Notably, our approach also leads in terms of the κ, achieving the highest value at 92.86%. Fig. 11 illustrates the classification maps generated by these diverse algorithms for a visual comparison. Our classification map

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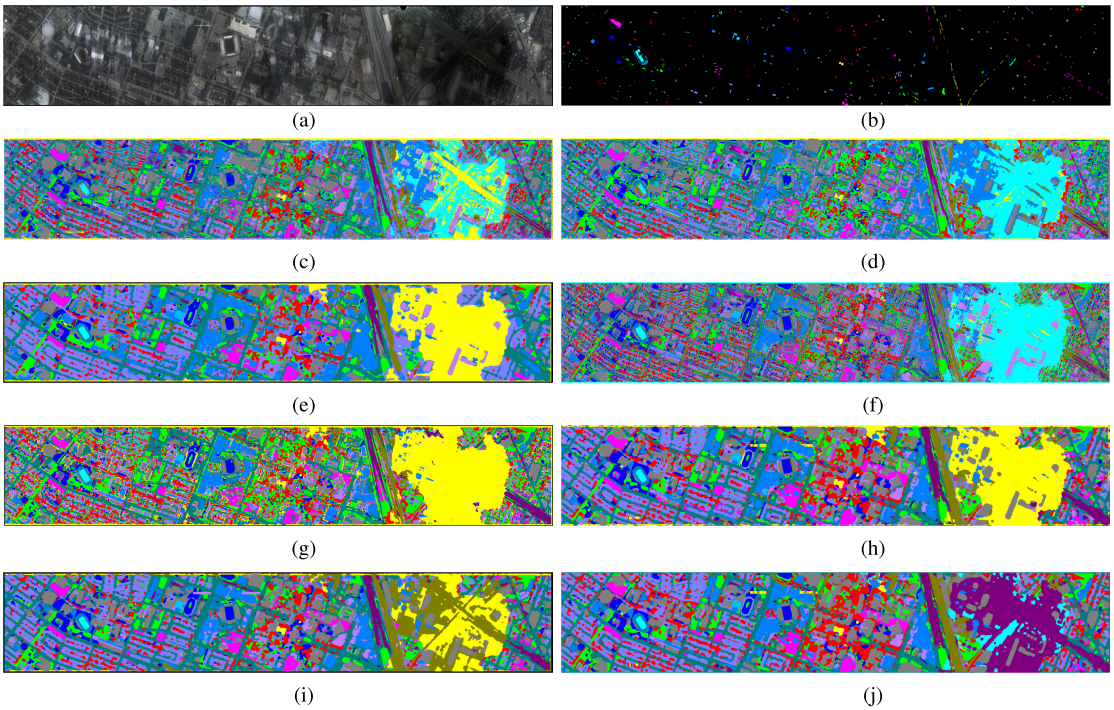


Fig. 9. Classification map for H13Train→H13Test task. (a) False-color image. (b) Ground truth. (c) DAN. (d) DANN. (e) ED-DMM-UDA. (f) CDA. (g) CLDA. (h) TAADA. (i) SCLUDA. (j) MSDA.

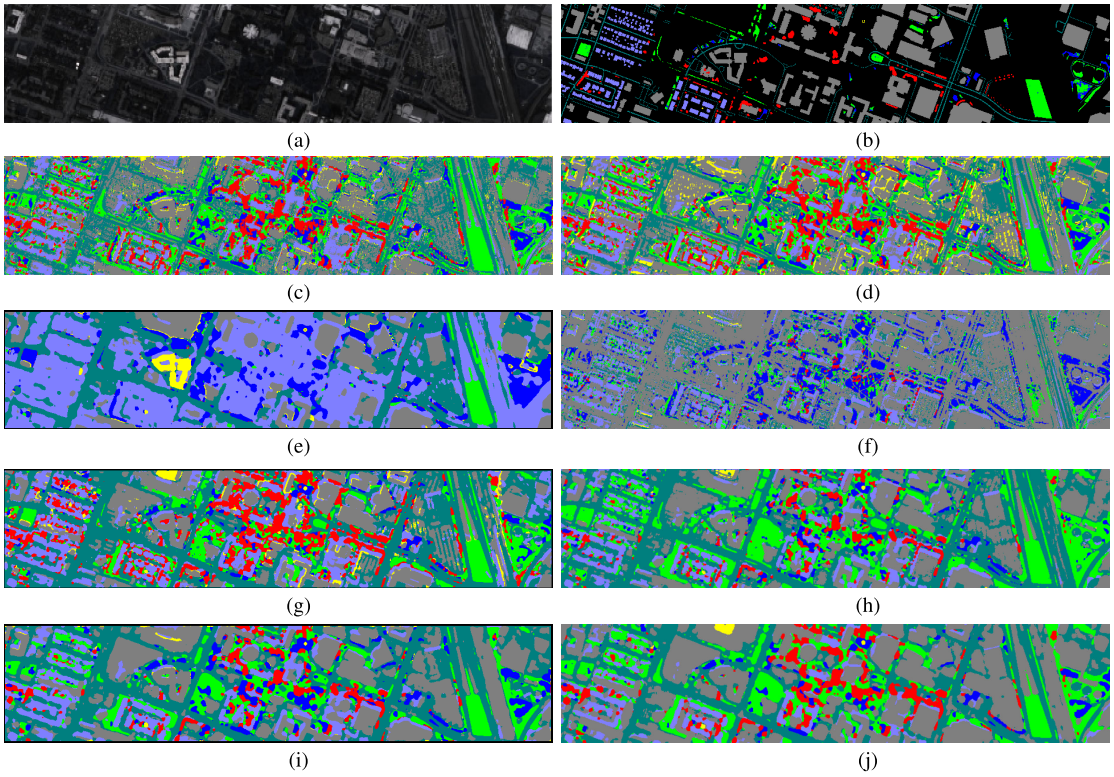


Fig. 10. Classification map for H13→H18 task. (a) False-color image. (b) Ground truth. (c) DAN. (d) DANN. (e) ED-DMM-UDA. (f) CDA. (g) CLDA. (h) TAADA. (i) SCLUDA. (j) MSDA.

clearly distinguishes Bitumen from the adjacent trees, and it also provides a clearer distinction between buildings and their surrounding areas.

On the Dioni→LouKia task, Table IX succinctly summarizes the experimental outcomes for the various algorithms. Among these, DANN registers the lowest performance, with an accuracy of 42.86%. Close contenders DAN, CDA, and CLDA each record accuracies in the vicinity of 50%. TAADA shows a modest improvement, achieving a classification accuracy of 56.26%. Notably, SCLUDA and ED-DMM-UDA make significant strides, attaining accuracies of 60.08% and 61.37%, respectively. Our method marks a further enhancement in this

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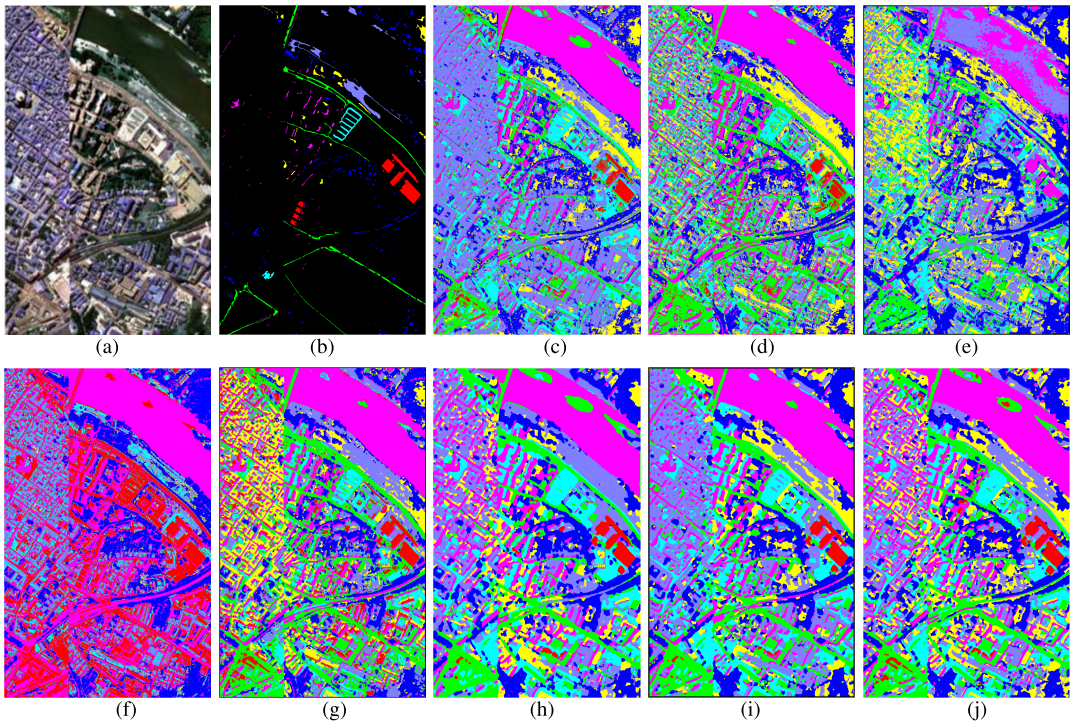


Fig. 11. Classification map for UP→PC task. (a) False-color image. (b) Ground truth. (c) DAN. (d) DANN. (e) ED-DMM-UDA. (f) CDA. (g) CLDA. (h) TAADA. (i) SCLUDA. (j) MSDA.

trend, culminating in a notable accuracy of 63.86% on the Dioni→LouKia task. Fig. 12 provides a visual representation of the classification maps generated by these diverse algorithms, offering an illustrative comparison of their performances. The separation between different types of seawater is clear, and the distinctions between various classes are both smooth and accurate.

For the NC12→NC13 task, the experimental results of various algorithms are presented in Table X. It is noteworthy that the target domain within this dataset is characterized by an imbalance, prominently featuring classes such as reed and water. Under these challenging circumstances, DAN, DANN, and ED-DMM-UDA exhibit limited success, demonstrating difficulty in effectively distinguishing categories, with an approximate accuracy of only 50%. CDA and SCLUDA show marked improvements, achieving accuracies close to 76%. Both CLDA and TAADA perform exceptionally, recording accuracies of 83.65% and 87.49%, respectively. However, our method surpasses these figures, achieving an impressive accuracy of 92.70%. In terms of AA, TAADA takes the lead with an accuracy of 75.79%, while our approach achieves the highest κ at 88.68%. Fig. 13 provides a visual depiction of the classification maps generated by these diverse algorithms, which is consistent with quantitative results. Our method provides a more reasonable differentiation between classes, achieving significant separability.

D. Ablation Study

To further validate the effectiveness of the CSAT scheme and MSD module, we perform ablation experiments on each part, and the results are shown in Table XI. In this ablation experiments, MCC denotes the utilization of the MCC loss, the absence of MCC signifies the exclusion of MCC loss from the CSAT scheme. MSD represents the incorporation of MSD, the absence of MSD implies the complete omission of masking strategy, self-distillation module, and MSD loss. Preserve central pixel (PCP) stands for preserving the central pixel of the target sample during the masking process. The absence of PCP indicates the central pixel is not intentionally retained in the MSD module. As shown in Table XI, the results of the ablation experiments demonstrate the effectiveness of each component of MSDA. Specifically, in the case of the CSAT scheme, the inclusion of the MCC loss leads to a notable improvement across all five tasks, except for a slight decrease on the Dioni→LouKia task. This indicates that incorporating MCC into the CSAT scheme can improve the performance of our method on almost all tasks. After the addition of MSD, the accuracy on each dataset increased by 1.13%, 1.7%, 2.13%, 6.1%, and 2.28%, respectively. This fully demonstrates the effectiveness of the MSD module. After the addition of PCP, the accuracy of each dataset also increased significantly. This proves that intentionally preserving the central pixel during the masking process is necessary.

In order to further illustrate the performance improvement of our method resulting from the addition of MSD, we utilize a Grad-CAM technique [64] to samples processed by models with mask, models without mask, and models trained solely on source samples to generate heatmaps for HSI. Specifically, Grad-CAM is applied to the last convolutional layer of the spectral attention module within the feature extractor to obtain spectral attention heatmaps. Likewise, the same technique is employed on the final layer of the spatial attention module to generate spatial attention heatmaps.

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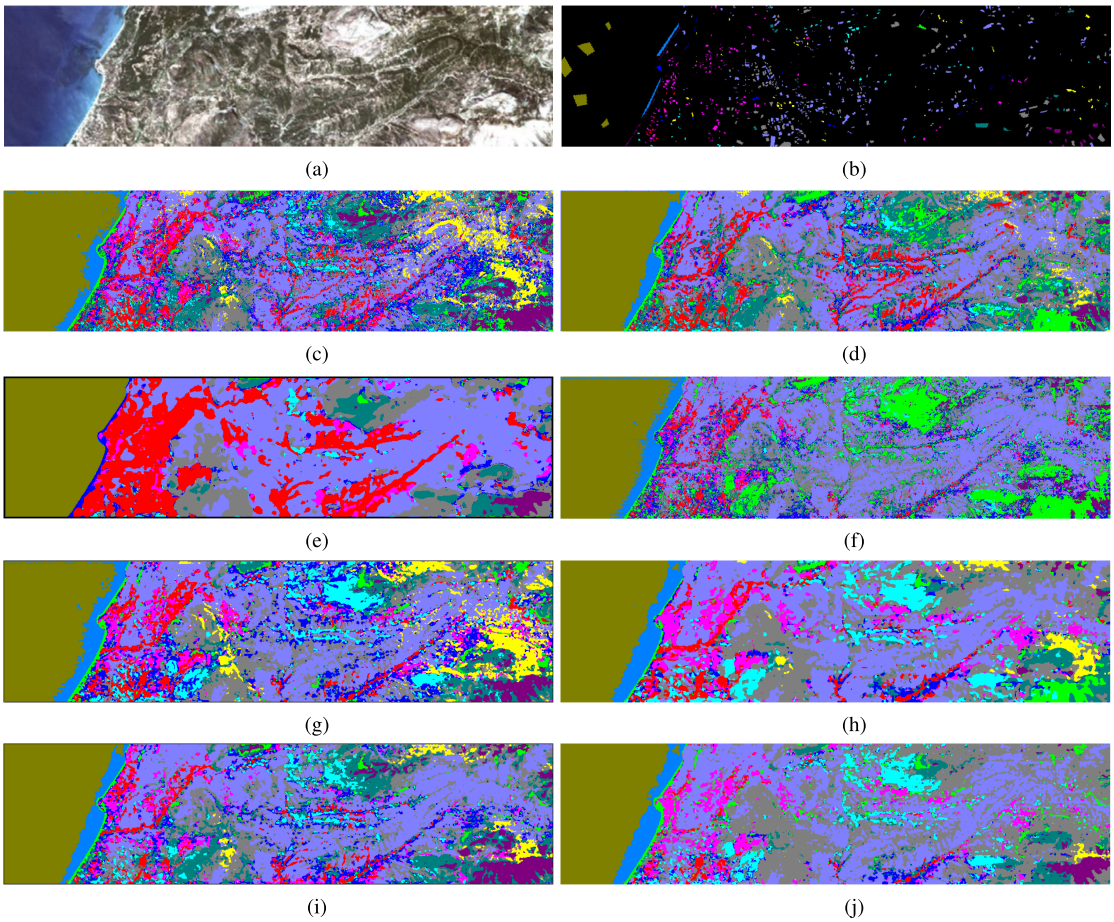


Fig. 12. Classification map for Dioni→LouKia task. (a) False-color image. (b) Ground truth. (c) DAN. (d) DANN. (e) ED-DMM-UDA. (f) CDA. (g) CLDA. (h) TAADA. (i) SCLUDA. (j) MSDA.

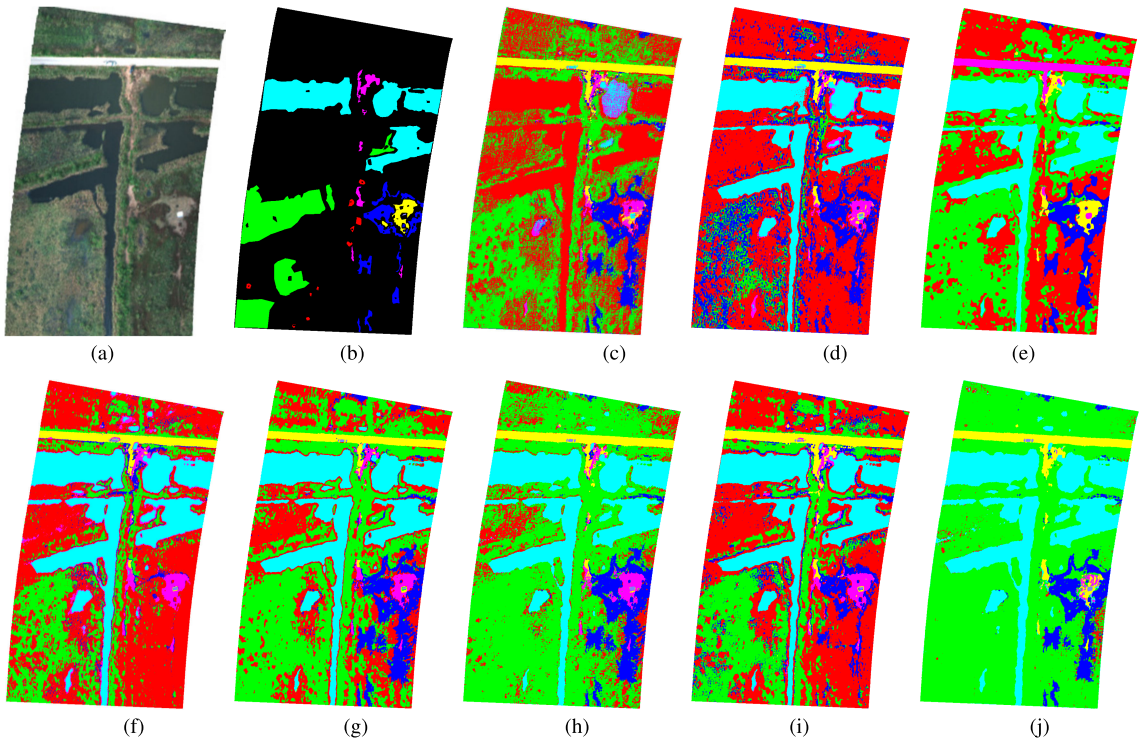


Fig. 13. Classification map for NC12→NC13 task. (a) False-color image. (b) Ground truth. (c) DAN. (d) DANN. (e) ED-DMM-UDA. (f) CDA. (g) CLDA. (h) TAADA. (i) SCLUDA. (j) MSDA.

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TABLE XI
ABLATION STUDY

No.	MCC	MSD	PCP	H13Train→H13Test	H13→H18	UP→PC	Dioni→LouKia	NC12→NC13
1	✘	✘	✘	86.87	71.85 ±9.57	90.91 ±1.35	54.54 ±6.12	87.77 ±2.22
2	✓	✘	✘	87.53	75.58 ±4.58	91.16 ±2.07	53.83 ±3.21	88.21 ±5.51
3	✓	✓	✘	88.45	77.28 ±2.36	93.29 ±1.00	60.64 ±1.79	90.05 ±4.90
4	✓	✓	✓	90.95	79.84 ±1.27	94.06 ±0.26	63.86 ±1.01	92.70 ±0.48

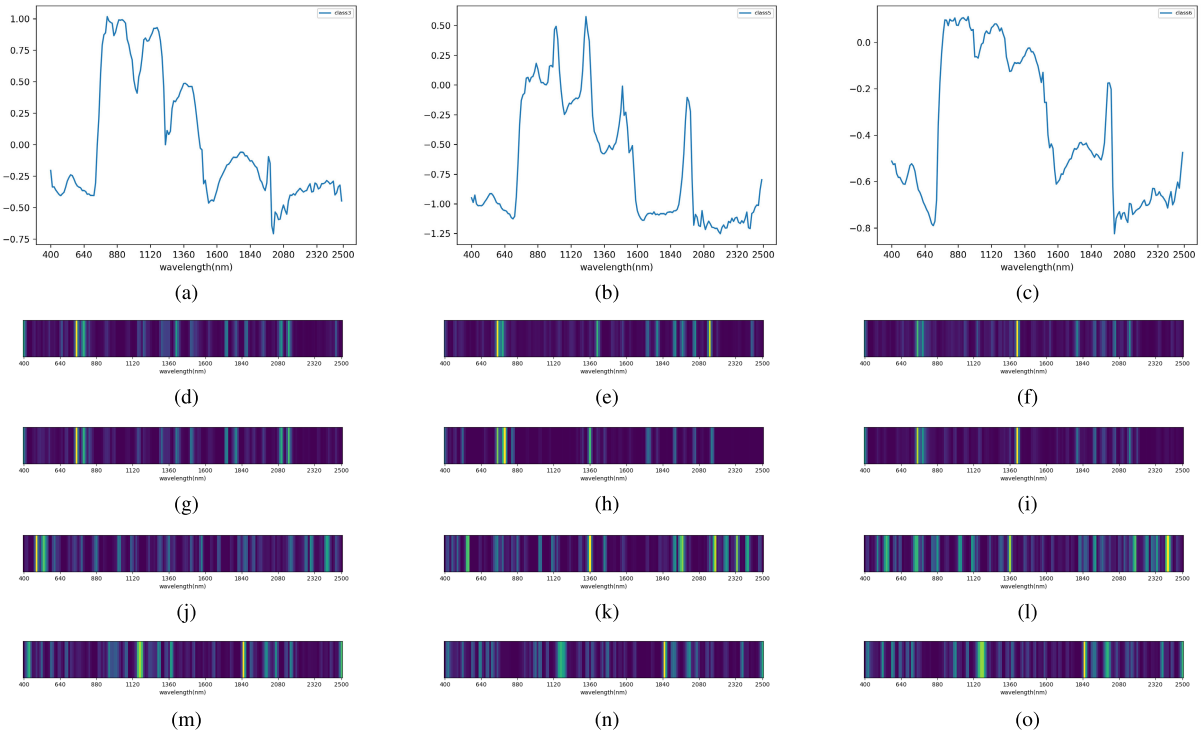


Fig. 14. Spectral band heatmaps on Dioni→LouKia task. Each column corresponds to a category of samples. (a)–(c) Average spectral curves of the source samples. (d)–(f) Heatmaps obtained from source samples in our MSDA method. (g)–(i) Heatmaps obtained from target samples in our MSDA method. (j)–(l) Heatmaps obtained from target samples in the method without MSD. (m)–(o) Heatmaps for target samples in the method trained only on source domain data without UDA.

The results of spectral attention are shown in Fig. 14, and the color gradient from yellow to blue indicates the score from high to low. We randomly select three categories, where the three columns from left to right represent "nonirrigated arable land," "olive groves," and "coniferous forest." Fig. 14(d) shows the model using source samples with emphasis on the 640–880-nm bands, underscoring the significance of these bands for the category. Fig. 14(g) shows our method's ability to similarly highlight the crucial 640–880-nm bands in the target domain. Fig. 14(j) indicates that using only domain alignment without masking leads the model to miss the sensitive band in the target domain, thus not recognizing the band-category relationship. Fig. 14(m) demonstrates that directly applying the source domain model to the target domain results in overlooking the important bands due to spectral drift. The heatmap outcomes confirm our model's effectiveness in focusing on the sensitive spectral bands. Lacking mask integration into target domain data, the model incorrectly focuses on nonsensitive bands.

From a spatial perspective, we analyze the differences between samples processed by the model with a mask and those without a mask. Three samples are randomly selected from the "grass stressed" category of the H18 dataset. As illustrated in Fig. 15, comparing Fig. 15(a) to (b), it becomes apparent that masking encourages the network to concentrate more on the intrinsic features of the classes and the pixels delineating interclass relationships. Our analysis indicates that the heatmaps produced by models utilizing masks demonstrate enhanced accuracy, with a more focused delineation of class boundaries compared to models without mask integration. The experimental data clearly show that the inclusion of our masking strategy significantly boosts the overall performance.

E. Analysis of Mask Strategies and Teacher Networks

In our experiments, we mask the HSI patch sample and feed it into the student networks, while the unmasked sample is passed through the teacher networks. By comparing the

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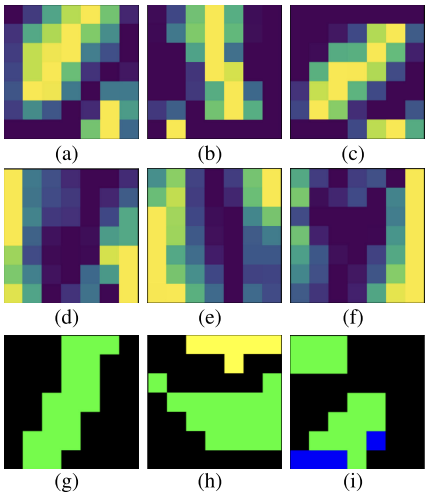


Fig. 15. Spatial heatmaps. We present spatial attention maps for three samples in the H13→H18 task, and each column corresponds to a sample. (a)–(c) Heatmaps obtained from the target samples using MSD. (d)–(f) Heatmaps obtained using the method without MSD. (g)–(i) Ground-truth labels of the sample.

TABLE XII
COMPARISON OF DIFFERENT MASK STRATEGIES AND TEACHER NETWORK STRUCTURES

Mask.	One classifier teacher	Biclassifier teacher
spectral	74.79	74.38
spatial	67.80	33.61
spectral + spatial	79.84	76.25

outputs of the two networks, we aim to minimize the difference between their outputs to achieve self-distillation. Here, we explore the strategies of masking and how to construct a teacher network.

1) Analysis of Mask Strategies: For HSI patch samples, we construct experiments with three mask strategies, including spatial mask, spectral mask, and spatial–spectral mask.

Spatial mask involves randomly masking several pixels based on the mask ratio at the pixel level. For the masked pixels, all spectral data of those pixels are masked.

Spectral mask involves randomly selecting several channels based on the mask ratio and masking all information of the selected channels in the sample.

Spatial–spectral mask, as mentioned earlier, involves masking at the snippet level.

The experimental results of different mask strategies on the H13→H18 task are presented in Table XII. Given the rich spectral information, it can be observed that employing only a spatial mask significantly reduces the network's performance. Similarly, using solely a spectral mask makes it challenging for the network to understand the relations between different pixels within the same band. Unlike the previous two strategies, a combined approach of spatial and spectral masking yields superior results. It can be observed that using the spatial–spectral mask strategy to simultaneously learn the class-relevant knowledge from both spatial and spectral dimensions can enhance the performance of the model.

2) Analysis of Teacher Networks: In our backbone networks, we utilized a feature extractor and two classifiers. For constructing the self-distillation networks, two approaches are considered. The first approach involved using G and C_1 to build single classifier teacher networks G_{tea} and $C_{1,\text{tea}}$, which are adopted in our MSDA method. The second approach utilized G, C_1, and C_2 to construct biclassifier teacher networks G_{tea}, $C_{1,\text{tea}}$, and $C_{2,\text{tea}}$. In this setting, the masked data passes through the student networks, and the features extracted after G are separately fed into C_1 and C_2 to obtain predictions. The unmasked data goes through the teacher networks, and the features obtained after G_{tea} are used by $C_{1,\text{tea}}$ and $C_{2,\text{tea}}$. The objective is to maintain the consistency of the outputs of C_1 and $C_{1,\text{tea}}$, as well as the outputs of C_2 and $C_{2,\text{tea}}$.

The experimental results of the two approaches to constructing the teacher networks on H13→H18 are presented in Table XII. It can be observed that the single-classifier teacher networks tailored for CSAT improve the performance more effectively than the biclassifier teacher networks, and reduce the computational burden during the model training phase.

F. Parameter Analysis

The results from different hyperparameter settings for the five experiments are shown in Fig. 16.

In our experiments, the key hyperparameters include patch size ps, snippet height or width h, snippet spectral dimensions s, mask ratio r, and MCC temperature T. Initially, the ps is determined to control the spatial dimensions of the samples, with options ranging from [3, 5, 7, 9]. The mask ratio r, indicating the proportion of data to be masked within a target sample, is selectable from a range of $[0.1, 0.2, \ldots, 0.9]$. The spectral dimensions of snippets, denoted by s, are critical for controlling the spectral range of the snippets, ensuring that the spectral bands within each snippet are closely correlated. The choices for s span from $[1, 2, \ldots, 15]$. The parameter h governs the length and width of each snippet, thereby influencing the spatial information. h is selected from $[1, 2, \ldots, \lceil \text{ps}/2 \rceil]$. The temperature parameter T, essential for probability rescaling, can be adjusted within $[1, 2, \ldots, 10]$.

In our experimental process, we first determine the ps for the samples. We then establish the mask ratio r to determine how much of the data is to be masked. Next, we choose s to ensure that the snippets have strongly correlated spectral bands and then determine h to control the spatial information of the snippet. Finally, we determine the parameters T for probability rescaling.

The results in Fig. 16 demonstrate that the setup of hyperparameters in the above experiments is appropriate, having a favorable impact on the performance of different domain adaptation tasks.

G. Computational Complexity

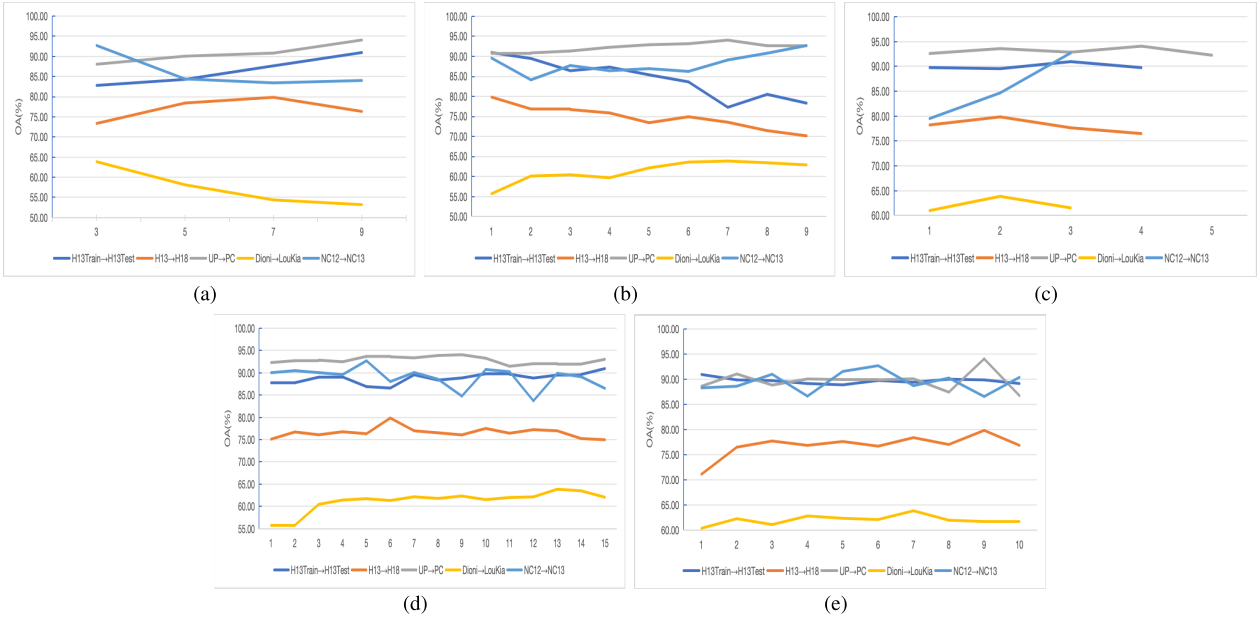
In Table XIII, we present a comprehensive comparison of computational complexities, including training time, testing time, floating-point operations (FLOPS), and number of parameters. All experiments were conducted on a computer equipped with a 2.9 GHz Intel[1] Xeon[1] Gold 6226R CPU and

[1]Registered trademark.

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TABLE XIII
COMPUTATIONAL TIME (SECONDS), FLOPS, AND PARAMETERS ON EACH TASK WITH DIFFERENT METHODS

	Methods	DAN	DANN	ED-DMM-UDA	CDA	CLDA	TAADA	SCLUDA	MSDA
H13Train→H13Test Testting time	Training time	485.93	42.34	1231.05	303.84	587.74	391.84	357.58	453.31
	1.42	1.58	10.29	0.07	0.74	1.14	1.28	0.98	
	FLOPS	18314767	27081296	12366602	26304515	12381478	69979841	76294730	69979841
	#params	26006903	13465040	189474	102765	204222	728624	1978417	728624
H13→H18	Training time	664.75	28.59	941.54	207.86	263.13	205.63	135.27	189.49
	Testting time	5.33	5.89	14.82	0.68	2.92	2.67	4.56	3.85
	FLOPS	4606599	15873800	6555402	19648507	6569238	7678089	76788089	15320609
	#params	25701743	13159880	170274	76757	183982	577640	577640	577640
UP→PC	Training time	648.98	31.81	303.2	201.21	465.15	283.82	331.25	428.09
	Testting time	4.61	5.77	13.72	0.47	2.18	3.83	4.44	3.3
	FLOPS	15017863	23784392	11686202	23104507	11700038	50180201	80408762	50180201
	#params	25871087	13329224	181074	90257	194782	661880	1876017	661880
Dioni→LouKia	Training time	459.12	42.49	1142.97	383.59	224.85	211.98	327.11	332.01
	Testting time	1.03	1.13	10.09	0.06	0.51	0.7	0.87	0.74
	FLOPS	3655052	15227149	1465802	28160512	1480288	29503089	80408762	9386105
	#params	26105716	13563853	195874	110012	210232	811028	1876017	778085
NC12→NC13	Training time	405.33	47.56	1479.39	310.71	124.66	156.82	837.48	210.16
	Testting time	19.98	22.31	58.76	1.55	13.17	13.65	16.83	16.09
	FLOPS	4831623	16403720	1635002	33856507	1648838	29503089	80408762	14309825
	#params	26397935	13856072	214674	132257	228382	811028	1876017	923960



(a) (b) (c) (d) (e)

Fig. 16. Sensitivity analysis of parameters using Houston dataset. (a) ps, (b) r, (c) h, (d) s, and (e) T.

an NVIDIA GeForce RTX 3090 GPU, using the open-source deep learning framework PyTorch, and the experimental environment is available in a docker image.[2] The computational cost of the proposed MSDA in the training process is higher than most of the other algorithms, but its testing process is faster than the majority of them.

V. CONCLUSION

In this article, we proposed a novel MSDA framework that integrates MSD into domain adaptation. MSD demonstrates

[2]docker address: https://hub.docker.com/r/he97/hsi_exp_env/tags.

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its powerful knowledge extraction capabilities on unlabeled target data, resulting in discriminative features that significantly enhance the inference accuracy of HSI classification. Our results indicate that UDA performance can be improved not only through distribution alignment but also by fully exploiting the target domain. Nevertheless, MSD only focuses on semantics inherent in the target domain samples and grows the computational complexity due to the introduction of self-distillation. Future research could consider techniques to improve computational efficiency and explore additional strategies tailored to the specific characteristics of unlabeled HSI data in the target domain, such as contrastive learning and other self-supervised learning techniques, for further improving the performance of cross-scene HSI classification.

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